SCHEDULER ARCADE: ARCHITECTURE & ENGINE

SIMULATION ENGINE (engine.js)
- Discrete integer clock ticks
- 5 core scheduling algorithms
- Exact context-switch penalty freeze

deterministic math

drives canvas frames

queries runtime state

EXPLAINABILITY TOOLS
- Live Kernel Decision Log terminal
- Real-time queue badges: MIN, VIP, AGED
- Process Diagnostic Inspector modal

viva key!

PIXEL RENDERER (ui.js)
- HTML5 Canvas 2D at 60 FPS
- Zero external image downloads
- Physical CPU Core socket display

triggers audio events

WEB AUDIO ENGINE (audio.js)
- Procedural square & triangle waves
- Synthesized 8-bit sound effects
- Zero audio media files

feeds workloads into

ARCADE MODES & BENCHMARKS
- 8-mission puzzle campaign
- Split-screen real-time versus duel
- 25/25 automated unit tests pass ☆